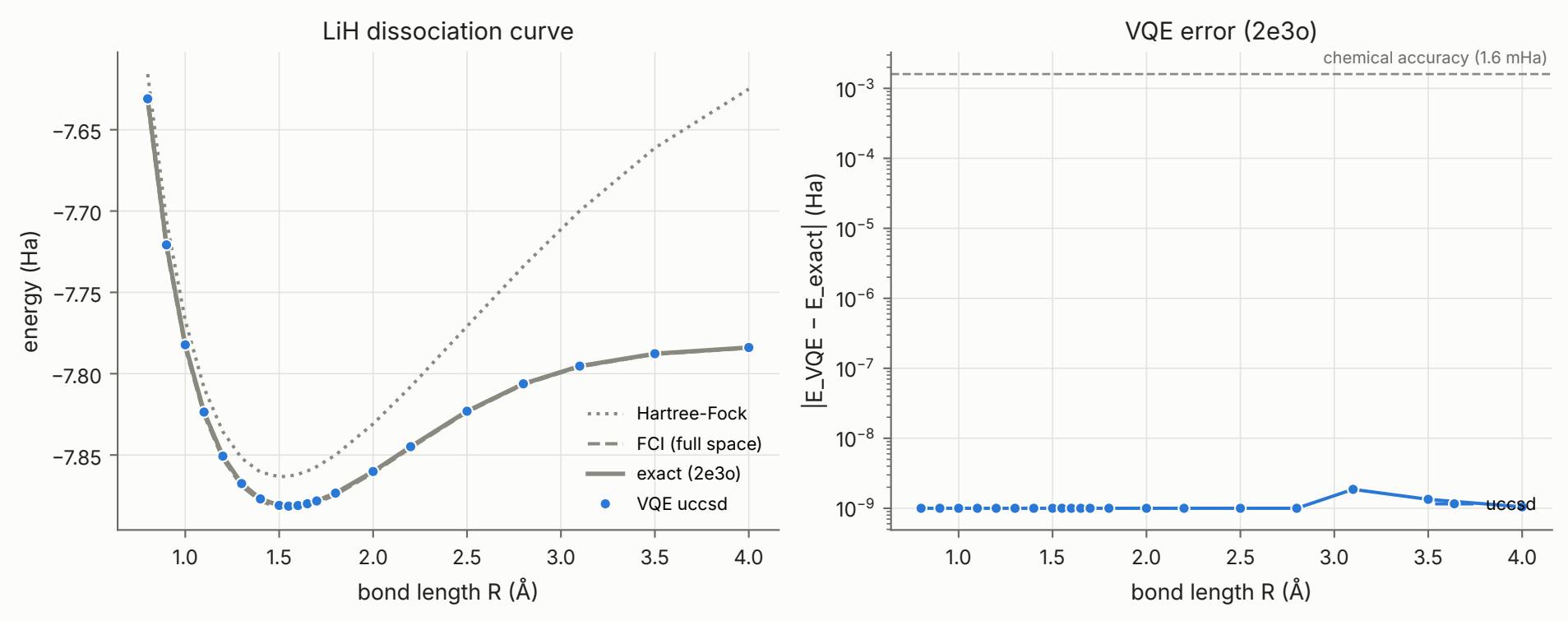
LiH dissociation curve
energy (Ha)
−7.65
−7.70
−7.75
−7.80
−7.85
1.0
1.5
2.0
2.5
3.0
3.5
4.0
bond length R (Å)
Hartree-Fock
FCI (full space)
exact (2e3o)
VQE uccsd
VQE error (2e3o)
chemical accuracy (1.6 mHa)
|E_VQE − E_exact| (Ha)
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
uccsd
1.0
1.5
2.0
2.5
3.0
3.5
4.0
bond length R (Å)